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On the Hyperboloid Distribution

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ABSTRACT. Through a group theoretic set up, using actions and invariant measures, the hyperboloid distribution is introduced as a very 'natural' probability measure on the unit-hyperboloid. This set up also brings forward a close analogy to the von Mises–Fisher distribution, the mathematics behind the two distributions being almost the same. Further, relevant questions of inference for the hyperboloid distribution are easily solved through the set up. At the end of the paper the theory developed is illustrated by a data example.

Key words: action; hyperboloid distribution; invariant measures; unithyperboloid; Von Mises–Fisher distribution

1. Introduction

It was pointed out in Barndorff-Nielsen (1978*b*) that the class of $(d-1)$-dimensional generalised hyperbolic distributions contains a subclass of distributions which in a natural way may be interpreted as a family of distributions on the unit-hyperboloid in $\mathbf{R}^d$, analogous to the family of von Mises–Fisher distributions on the unit sphere in $\mathbf{R}^d$. The distributions in question are called *hyperbola distributions* when $d=2$ and *hyperboloid distributions* for general d. The analogy between the hyperbola distribution and the von Mises distribution is very extensive, as has been demonstrated by Rukhin (1974) and Barndorff-Nielsen (1978*b*). It will be shown here that the analogy is equally extensive for general d and that in certain respects the hyperboloid distributions are even more tractable than the von Mises–Fisher distributions. The basis for the investigation is an analogue $x \ast y = x_1y_1 - x_2y_2 - \ldots - x_dy_d$ of the usual inner product $x \cdot y = x_1y_1 + \ldots + x_dy_d$ of two d-dimensional vectors x and y together with a related group of transformations of $\mathbf{R}^d$ which leaves the unit-hyperboloid H_d invariant and acts transitively on H_d.

The hyperboloid distributions may also be viewed as distributions for observational pairs (u, v) where u is a positive variate and v is a $(d-1)$-dimensional unit vector. The conditional distribution of v given u is then a von Mises–Fisher distribution. In particular, for $d=3$ the family of hyperboloid distributions provides a model for a simultaneous record of a linear and an angular variable. Such data one meets for instance in the Earth Sciences. An example is furnished by speed and direction of the wind, and a set of wind data of this kind will be considered in conjunction with the hyperboloid model in section 4. Not much statistical work have previously been done within the area of models for jointly observed linear and angular variates. Two papers should be mentioned in this connection. Mardia & Sutton (1978) present a model, where the marginal distribution of the angle is von Mises and the conditional distribution of the linear variable given the angle is normal, and they construct a test of independence. Johnson & Wehrly (1978) derive several models by requiring either maximum entropy under certain conditions or specified marginals. However, none of these models have any obvious nice mathematical structure, so inference may be difficult to draw.

The unit-hyperboloid in its group-theoretic setting is discussed in section 2. In section 3 the hyperboloid distribution is introduced and the problems of sample theory, estimation and testing are considered. Finally, section 4 gives an example of the use of the hyperboloid distribution.

2. The unit-hyperboloid in $\mathbf{R}^d$ and related topics

In this section we will discuss the unit-hyperboloid in d-dimensional euclidean space $\mathbf{R}^d$. A group of transformations on $\mathbf{R}^d$, analogous to the orthogonal group, and a measure on the unit-hyperboloid, invariant under these transformations, will be given.

For the notions of action and invariant measure we refer to Bourbaki (1960, 1963) and Nachbin (1965). A short outline may be found in Anderson (1978).

The results presented in this section can to a great extent be found also in Vilenkin (1968).

A. *A bilinear symmetric form and its 'unit sphere'*
If x is a vector in $\mathbf{R}^d$ we will think of x as a column-vector and denote the transpose of x by x'. We introduce a parallel to the usual inner product by

the operation $\ast$. For vectors x, y in $\mathbf{R}^d$, $x=(x_1, ..., x_d)'$, $y=(y_1, ..., y_d)'$, we define

$$x \ast y = x_1 y_1 - x_2 y_2 - ... - x_d y_d$$

which is a bilinear symmetric form. From the $\ast$-product we get a signed length in $\mathbf{R}^d$ defined by

$$R(x) = \operatorname{sign}(x \ast x)\sqrt{|x \ast x|}.$$

Moreover, the 'unit sphere' with respect to $\ast$ is defined as

$$H_d = \{x=(x_1, ..., x_d)' \in \mathbf{R}^d \mid x_1 > 0,\ R(x)=1\}.$$

Since the sets $\{x \in \mathbf{R}^d \mid x_1 > 0,\ R(x)=r\}$ are hyperboloids we shall call H_d the unit-hyperboloid. However H_d is termed the pseudosphere in Vilenkin (1968).

The interpretation of R is facilitated by introducing hyperbolic-spherical coordinates for the region Ω_d defined as

$$\Omega_d = \{x \in \mathbf{R}^d \mid x_1 > 0,\ R(x) > 0\}.$$

The coordinates are determined by:

$d=2$

$$x_1 = r\cosh u$$
$$x_2 = r\sinh u$$

and $r>0$, $u \in \mathbf{R}$.

$$d>2 \tag{1}$$

$$\begin{aligned} x_1 &= r\cosh u \\ x_2 &= r\sinh u \cos v_1 \\ x_3 &= r\sinh u \sin v_1 \cos v_2 \\ &\vdots \\ x_{d-1} &= r\sinh u \sin v_1 \sin v_2 \ldots \sin v_{d-3} \cos v_{d-2} \\ x_d &= r\sinh u \sin v_1 \sin v_2 \ldots \sin v_{d-3} \sin v_{d-2} \end{aligned}$$

and $r>0$, $u>0$, $v_1, ..., v_{d-3} \in (0, \pi)$, $v_{d-2} \in (0, 2\pi)\setminus\{\pi\}$, where r, of course, is $R(x)$. In the following $R(x)$ will be called the *hyperbolic length* and $x/R(x)$ the *hyperbolic direction.*

If we denote the transformation of $x=(x_1, ..., x_d)'$ into $(r, u, v_1 ..., v_{d-2})$ by h then the Jacobian of h^{-1} is

$$J(h^{-1}) = r^{d-1}\sinh^{d-2}u \sin^{d-3}v_1 \ldots \sin v_{d-3}. \tag{2}$$

B. *The group of hyperbolic transformations*

The special orthogonal group $S0(d)$ of transformations in $\mathbf{R}^d$ is defined by

$$\mathrm{S0}(d) = \{A,\ d \times d \text{ matrix} \mid A'A = I_d,\ \det A = 1\}$$

and has the main property that $(Ax)\cdot(Ay) = x \cdot y$ for vectors x, y in $\mathbf{R}^d$, where the dot denotes the usual inner product; this property implies that the unit sphere is left invariant under transformations from $S0(d)$.

We shall now introduce an analogous group relative to the $\ast$-product. Let $\tilde{I}_d$ be the $d \times d$ matrix

$$\tilde{I}_d = \begin{bmatrix} 1 & & & 0 \\ & -1 & & \\ & & \ddots & \\ & & & -1 \end{bmatrix}.$$

We define

$$SH(d) = \{A,\ d \times d \text{ matrix} \mid A'\tilde{I}_d A = \tilde{I}_d,\ \det A = 1, \\ \text{and } A_{11} > 0\}.$$

Here A_{11} means the (1,1)'th element of A.

If $A=(a_1, ..., a_d)$, where a_i is the ith column in A, the property $A'\tilde{I}_d A = \tilde{I}_s$ is equivalent to

$$R(a_i) = \begin{cases} 1 & i=1 \\ -1 & i=1, ..., d \end{cases} \quad \text{and} \quad a_i \ast a_j = 0,\ i \neq j. \tag{3}$$

The elements of $SH(d)$ will be called the hyperbolic transformations of $\mathbf{R}^d$.

Lemma 1. *We have*

(i) *The $\ast$-product is invariant under hyperbolic transformations, i.e. for $A \in SH(d)$ and vectors $x, y \in \mathbf{R}^d$ we have $(Ax)\ast(Ay) = x \ast y$.*

(ii) *The unit-hyperboloid is invariant under hyperbolic transformations, i.e. for $A \in SH(d)$ we have $A(H_d) \subseteq H_d$.*

(iii) *($SH(d)$, matrix multiplication) is a group where the inverse of A is $\tilde{I}_d A' \tilde{I}_d$.* □

Proof. That the $\ast$-product is invariant follows from the fact that $x \ast y = x'\tilde{I}_d y$ for vectors $x, y \in \mathbf{R}^d$.

To prove the invariance of H_d we must show that for $A \in SH(d)$ and $x \in H_d$ the first element of Ax is positive, $(Ax)_1 > 0$. First note that for vectors x, y in H_d we have that $x \ast y \geq 1$ with equality if and only if $x=y$, as may be seen from the hyperbolic-spherical representation (1) and the fact that $\cosh^2 u - \sinh^2 u = 1$. Next let $\zeta = (1, 0, ..., 0)'$ and $x \in SH(d)$. Then $(A\zeta)\ast(Ax) = \zeta \ast x = x_1 > 0$. Since either Ax or $-Ax$ belongs to H_d (from (i)) and $A\zeta \in H_d$ (from (3) and the definition of $SH(d)$ it is seen that the first column of A belongs to H_d) we find from the first remark that $Ax \in H_d$.

It is now trivial to prove the remaining parts of Lemma 1 using the definition of $SH(d)$. □

Next we give some examples of hyperbolic transformations:

$d=2$

$$SH(2)=\left\{\begin{bmatrix}\cosh\chi & \sinh\chi\\ \sinh\chi & \cosh\chi\end{bmatrix}\,\middle|\,\chi\in\mathbf{R}\right\}$$

$d>2$

$$\begin{bmatrix}1 & 0\dots0\\ 0 & \\ \vdots & B_1\\ 0 & \end{bmatrix}\quad \text{where } B_1'B_1=I_{d-1} \text{ and } \det B_1=1 \tag{4 a}$$

$$\begin{bmatrix}\cosh\chi & \sinh\chi & 0\dots0\\ \sinh\chi & \cosh\chi & 0\dots0\\ 0 & 0 & \\ \vdots & \vdots & B_2\\ 0 & 0 & \end{bmatrix}$$

$$\text{where } B_2'B_2=I_{d-2} \text{ and } \det B_2=1. \tag{4 b}$$

The group $SH(4)$ is the so-called restricted Lorentz group, well known from mathematics and physics. The Lorentz-group is defined as $\{A,\ 4\times4 \text{ matrix}\,|\,A'\tilde{I}_4A=\tilde{I}_4\}$ (Päerl, 1969). In relativistic mechanics one considers the subgroup of $SH(4)$ defined by

$$\left\{\begin{bmatrix}\cosh\chi & 0 & 0 & \sinh\chi\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ \sinh\chi & 0 & 0 & \cosh\chi\end{bmatrix}\,\middle|\,\chi\in\mathbf{R}\right\};$$

this is called the group of pure Lorentz transformations, and it is seen that the group is isomorphic to $SH(2)$. Generally the group $SH(d)$ may be termed the restricted pseudo-orthogonal group of order $(1,d-1)$, the pseudo-orthogonal group of order $(1,d-1)$, $O(1,d-1)$, being defined as

$$\{A,\ d\times d \text{ matrix}\,|\,A'\tilde{I}_dA=\tilde{I}_d\},$$

which is known from mathematics.

It follows from Lemma 1 that a hyperbolic transformation restricted to the unit-hyperboloid H_d, respectively $\Omega_d=\{x\in\mathbf{R}^d\,|\,x_1>0,\ \ R(x)>0\}$, is a bijective function on H_d, respecively Ω_d. In other words, the group of hyperbolic transformations $SH(d)$ acts on H_d, respectively Ω_d. Furthermore $SH(d)$ is a topological group since it is a subgroup of the general linear group.

Lemma 2. *We have*

(i) *$SH(d)$ and H_d are σ-compact and locally compact.*
(ii) *$SH(d)$ acts properly and transitively on H_d.* □

Proof. Since $SH(d)$ and H_d are closed subsets of $\mathbf{R}^{d^2}$ and $\mathbf{R}^d$, respectively, the first statement is obvious.

The transitivity follows if, for every vector a in H_d, we can find a matrix in $SH(d)$ such that a is the first column in this matrix. That this is indeed possible may be seen by composing the two matrices in (4).

Finally, to show that we have a proper action we look at

$$SH(d)\times H_d\to H_d\times H_d$$
$$(A,x)\to(Ax,x)$$

which is clearly a continuous function. We must show that it is moreover proper, i.e. that the inverse image of a compact set is compact. If (Ax,x) belongs to a compact set then x and Ax both belong to bounded sets. Let a be the first row in A. Since $\tilde{I}_dA'\tilde{I}_d=A^{-1}\in SH(d)$ and the first column of an element in $SH(d)$ belongs to H_d (see the proof of Lemma 1) we find that $\tilde{I}_da\in H_d$. Now $(Ax)_1=(\tilde{I}_da)\ast x$ is bounded so a itself is bounded. Finally, since the columns in A have R-lengths ±1, cf. (3), we see that the columns too are bounded vectors. □

If we generalise $\ast$ to

$$x\ast y=x_1y_1+\dots+x_ky_k-x_{k+1}y_{k+1}-\dots-x_dy_d,$$

for some $k=1,2,\dots,d$, and introduce a 'unit sphere' H_d^k and a 'special orthogonal group' SH_d^k relative to $\ast$ (in fact, for $1<k\leqslant d$ this is the special pseudo-orthogonal group of order $(k,d-k)$, denoted $SO(k,d-k)$), we find that SH_d^k acts transitively on H_d^k for all k. However, the action is proper only for $k=1$ and $k=d$. This may be seen as follows. Let $1<k<d$ and let γ be the action

$$SH_d^k\times H_d^k\xrightarrow{\ \gamma\ }H_d^k\times H_d^k$$
$$(A,x)\to(Ax,x)$$

then

$$\gamma^{-1}(\{(1,0,\dots,0)'\times(1,0,\dots,0)'\})$$

$$\supseteq\left\{\begin{bmatrix}1 & 0 & \dots & & & 0\\ 0 & \cosh\chi & 0\dots0 & \sinh\chi & 0\dots0\\ & 0 & 1 & 0 & & \\ & \vdots & \ddots & \vdots & \vdots & \vdots\\ & 0 & 1 & 0 & & \\ \vdots & \sinh\chi & 0\dots0 & \cosh\chi & 0\dots0\\ & 0 & & 0 & 1 & 0\\ & \vdots & & \vdots & \ddots & \\ 0 & 0 & & 0 & 0 & 1\end{bmatrix}\,\middle|\,\chi\in\mathbf{R}\right\}$$

$$\times\{1,0,\dots,0)'\}$$

which is not a bounded set.

It may, moreover, be noted that by choosing an appropriate basis for $\mathbf{R}^d$ every bilinear symmetric functional can be reduced to the form given by the generalised $\ast$-product.

We shall not investigate the cases $k=2, ..., d-1$ any further here, because an attempt to generalise the density, which we shall introduce in the following for $k=1$ (see (6)), gives a density that is not integrable, i.e. for $1<k<d$ there is no probability distribution of the form given by (6) below.

C. *An invariant measure on the unit-hyperboloid* H_d

We now want to find an invariant measure on the unit-hyperboloid H_d. Let λ_d be the restriction of the Lebesgue measure in $\mathbf{R}^d$ to the set Ω_d. Since the transformation $x \to Ax$ has determinant 1 for $A \in SH(d)$, we see that λ_d is invariant under hyperbolic transformations. Now, the mapping

$$\Omega_d \to H_d$$
$$v \to \frac{v}{R(v)}$$

is equivariant, and

$$\Omega_d \to \mathbf{R}$$
$$v \to R = R(v)$$

is invariant, so that the lifted measure $(V/R, R)(\lambda_d)$ is a product measure, $(V/R, R)(\lambda_d) = \nu \times \beta$, where ν is an invariant measure on the unit-hyperboloid H_d (cf., for instance, Barndorff-Nielsen et al., 1980).

Letting $\mathcal{B}_{H_d}$ be the σ-algebra generated by the compact sets in H_d, we define for $C \in \mathcal{B}_{H_d}$

$$\mu_d(C) = d \cdot \lambda_d\left(\left\{x \in \Omega_d \,|\, 0 < R(x) \leqslant 1, \ \frac{x}{R(x)} \in C\right\}\right).$$

From the above considerations it follows that μ_d is an invariant measure on the unit-hyperboloid H_d. From (2) it is seen that β, in the representation $(V/R, R)(\lambda_d) = \mu_d \times \beta$, is given by $d\beta(r) = r^{d-1}dr$ and that the measure μ_d in hyperbolic-spherical coordinates is given by

$$\sinh^{d-2} u \sin^{d-3} v_1 \ldots \sin v_{d-3} du\, dv_1 \ldots dv_{d-2}. \tag{5}$$

3. The hyperboloid distribution and its statistical properties

As mentioned in the introduction the hyperboloid distribution is a distribution on the unit-hyperboloid, which is analogous to the von Mises–Fisher distribution on the unit-sphere. Essentially, the only difference is that we use the structure on $\mathbf{R}^d$ given by the $\ast$-product instead of the usual inner product. In the present section we first introduce the distribution and then investigate some of its statistical properties, drawing on the analogy with the von Mises–Fisher distribution.

A. *The distribution*

We define a probability measure $P_{\xi,\varkappa}$ on the unit-hyperboloid, absolutely continuous relative to the invariant measure μ_d and with parameters ξ and $\varkappa$, ξ belonging to the unit-hyperboloid H_d and $\varkappa$ positive real:

$$\frac{dP_{\xi,\varkappa}}{d\mu_d}(v) = a_d(\varkappa)\, e^{-\varkappa \xi \ast v}, \ v \in H_d. \tag{6}$$

The family of these distributions is called the family of $(d-1)$-dimensional *hyperboloid distribution* and we shall denote it by $\mathcal{D}$, i.e.

$$\mathcal{D} = \{P_{\xi,\varkappa} \,|\, \xi \in H_d, \ \varkappa > 0\}.$$

The parameters ξ and $\varkappa$ will be called, respectively, the direction and the concentration of the hyperboloid distribution.

For a hyperbolic transformation $A \in SH(d)$ we find from the invariance of the $\ast$-product, see Lemma 1, and the invariance of μ_d that the lifted measure $A(P_{\xi,\varkappa})$ is the measure $P_{A\xi,\varkappa}$, and that the norming constant depends on $\varkappa$ only. It also follows from this and the fact that $SH(d)$ acts transitively on H_d, see Lemma 2, that for fixed $\varkappa$ we have a group family, i.e. the family is generated by lifting a single member with the hyperbolic transformations; furthermore, this family is clearly exponential of order d.

The distribution (6) was introduced in Barndorff-Nielsen (1978) (though in a different set up) as a special case of the generalised hyperbolic distribution.

The distribution may be characterised in a number of ways. We shall not discuss this here, except for mentioning that for $d=2$ Rukhin (1974) has given a characterisation relating to the notion of strongly symmetrical distributions. Moreover, for $d=2$ the distribution (6) may be seen to be the distribution of the logarithm of a generalised inverse Gaussian variate with index parameter $\lambda=0$ (Barndorff-Nielsen, 1978*b*, see also Jørgensen, 1980).

It has long been known that the von Mises–Fisher distribution is obtainable from the normal distribution with identity variance matrix by conditioning on the observation falling on a sphere (Downs, 1966). Such a simple result does not hold generally for the distribution (6), but for $d=2$ or 3 it is possible to obtain the distribution (6) from the normal distribution by suitable conditioning, see Blæsild (1979).

If we use hyperbolic-spherical coordinates on the unithyperboloid

$$\xi = (\cosh\chi, \sinh\chi\cos\theta_1, \ldots, \sinh\chi\sin\theta_1\ldots\sin\theta_{d-2})',$$
$$= (\cosh\chi, \sinh\chi\, e(\theta))'$$

and

$$v = (\cosh u, \sinh u e(w))',$$

where $e(\theta)$ and $e(w)$ belong to the unit sphere $E_{d-1} = \{x \in \mathbf{R}^{d-1} | \|x\| = 1\}$, we find that (see (5))

$$dP_{\xi,\varkappa}(u, w_1, \ldots, w_{d-2})$$
$$= a_d(\varkappa)\sinh^{d-2} u \sin^{d-3} w_1 \ldots \sin w_{d-3}$$
$$\times \exp(-\varkappa\{\cosh\chi\cosh u - \sinh\chi\sinh u$$
$$\times (e(\theta)\cdot e(w))\})\, du\, dw_1 \ldots dw_{d-2}. \tag{7}$$

From this it follows that

$$a_d(\varkappa) = \frac{\varkappa^{d/2-1}}{(2\pi)^{d/2-1}\, 2K_{d/2-1}(\varkappa)} \tag{8}$$

where K_ν is the modified Bessel function of the third kind and with index parameter ν (see Barndorff-Nielsen, 1978*b*).

For two and three dimensions the formulas are as follows

$d = 2$

$$\frac{1}{2K_0(\varkappa)}\exp(-\varkappa\cosh(u-\chi))\, du, u \in \mathbf{R}.$$

$d = 3$

$$(\varkappa e^{\varkappa}/2\pi)\sinh u$$
$$\times \exp(-\varkappa(\cosh\chi\cosh u - \sinh\chi\sinh u\cos$$
$$(w-\theta)))\, du\, dw, \ u \geq 0, \ w \in [0, 2\pi).$$

From (7) it can be shown that the marginal density of u is asymptotically equivalent to

$$\text{const.} \exp((d-2)u)\exp(-(\varkappa/2)(\cosh\chi - \sinh\chi))\, e^u$$

as u tends to infinity. So, in the tail the distribution of u resembles an extreme value distribution.

For $d=2$ the variate u may be shown not to be infinitely divisible (the density dies off too quickly). Following Kent (1977) one can define a 'hyperboloid sum' on the unit-hyperboloid and ask whether or not a distribution on the unit-hyperboloid is infinitely divisible with respect to this sum. In two dimensions this reduces to the question of whether or not u is infinitely divisible and so the answer is negative for the hyperbola distribution.

From the form of the density in hyperbolic-spherical coordinates (7) it is seen that $e(w)$ *given u follows a von Mises–Fisher distribution on E_{d-1} with mean-direction $e(\theta)$ and concentration* $\varkappa\sinh\chi\sinh u$ and that *for $\xi = (1, 0, \ldots, 0)'$ the variate u and the direction $e(w)$ are independent, $e(w)$ being uniformly distributed on E_{d-1} while u has density* $\tilde{a}_d(\varkappa)\sinh^{d-2} u e^{-\varkappa\cosh u}$; in three dimensions, still for $\xi = (1, 0, \ldots, 0)'$, the cumulative distribution function for u is particularly simple, namely $1 - e^{\varkappa(1-\cosh u)}$ for $u > 0$, i.e. the variate $\cosh u - 1$ follows an exponential distribution with parameter $\varkappa$.

Finally, in this brief view of the distribution $P_{\xi,\varkappa}$ we state what happens as $\varkappa$ becomes very large or very small.

For $\varkappa \to \infty$ the distribution $P_{\xi,\varkappa}$ becomes concentrated around the parameter value ξ (which is in fact the hyperbolic direction of the mean of the distribution, see Lemma 3). To be more specific we let $\zeta = (1, 0, \ldots, 0)'$ and, using a Taylor expansion of $\cosh u$ and $\sinh u$, we find that under $P_{\xi,\varkappa}$ and for $\varkappa \to \infty$ the distribution of $\varkappa u^2$ tends to a chi-squared distribution with $d-1$ degrees of freedom and the distribution of $\sqrt{\varkappa}\sinh u \cdot e(w)$ tends to a $(d-1)$-dimensional normal distribution with expectation zero and identity variance matrix. For general $\xi = (\cosh\chi, \sinh\chi\, e(\theta))$ we find, by using a hyperbolic transformation and the above result for the case $\xi = \zeta$, that under $P_{\xi,\varkappa}$ and for $\varkappa \to \infty$ the distribution of the variate $\sqrt{\varkappa}(\cosh u - \cosh\chi)$ tends to a normal distribution with mean zero and variance $\sinh^2\chi$ while $\sqrt{\varkappa}$ times the projection of ξ minus the observation onto the tangent plane to H_d through ξ tends in distribution to a $(d-1)$-dimensional normal random variable with mean zero and variance matrix of the form

$$\begin{bmatrix} \cosh 2\chi & 0 \ldots 0 \\ 0 & 1 \\ \vdots & & \ddots \\ 0 & \ldots & 1 \end{bmatrix},$$

in a suitably chosen orthonormal basis for the tangent plane.

For $\varkappa \to 0$ the ratio between the density w.r.t. μ_d in any two points tends to unity, uniformly over bounded regions, i.e. the probability mass is spread out uniformly over the hyperboloid.

In the following parts many results are analogous to results for the von Mises–Fisher distribution and the proofs are quite similar. Accordingly, some of the proofs are omitted here. For the von Mises–Fisher case the reader is referred to Mardia (1972, 1975) and Schou (1978).

B. *Distribution theory for a single sample*

Let $v_1, \ldots, v_n$ be a sample from the distribution $P_{\xi,\varkappa}$. In the following we will consider

$$v. = \sum_1^n v_i$$

$$R = R(v.) = \sqrt{v. \ast v.}$$

$$T = \xi \ast v..$$

R will be termed the resultant length and $v./R$ the resultant direction. Then the joint distribution of $v_1, \ldots, v_n$ is

$$\frac{dP^n_{\xi,\varkappa}}{d\mu^n_d}(v_1, \ldots, v_n) = a_d(\varkappa)^n \exp(-\varkappa\xi \ast v.)$$

$$= a_d(\varkappa)^n \exp\left(-\varkappa R\xi \ast \frac{v.}{R}\right).$$

It is seen that $v.$ or equivalently, $(R, v./R)$ is minimal sufficient for the family $\mathcal{P}$.

An important tool for inference in the von Mises-Fisher distribution is that the length and direction of the resultant vector are independent under the uniform distribution on the sphere. A similar result is applicable here. We let $SH(d)$ act on H^n_d by acting in each coordinate separately. Then from Lemma 2 we have that $SH(d)$ acts properly on H^n_d. Furthermore, the mapping

$$\left(\frac{v.}{R}, R\right): H^n_d \to H_d \times \mathbf{R}_+$$

is proper, and this implies that

$$\left(\frac{v.}{R}, \pi\right): H^n_d \to H_d \times H^n_d/SH(d)$$

is proper, where

$$\pi: H^n_d \to H^n_d/SH(d)$$

$$(v_1, \ldots, v_n) \to SH(d)(v_1, \ldots, v_n)$$

is the orbit projection. It now follows, cf. for instance Barndorff-Nielsen et al. (1980), that

Theorem 1. *We have*

$$\left(\frac{v.}{R}, \pi\right)(\mu^n_d) = \mu_d \times \delta,$$

where δ is a measure on the orbit space $H^n_d/SH(d)$, and

$$\left(\frac{v.}{R}, R\right)(\mu^n_d) = \mu_d \times \psi^n_d,$$

where ψ^n_d is a measure on $\mathbf{R}_+$.

Moreover, the orbit projection π and the resultant direction $v./R$ are conditionally indepenent given the resultant length R, in symbols

$$\pi \perp \frac{v.}{R} \mid R. \qquad \square$$

We are now able to explore the structure of the distribution $P_{\xi,\varkappa}$ in order to get some guidance on how to draw the inference and to obtain the distributions of statistics used in the inference.

Theorem 2. *The distribution of the resultant length is given by*

$$\frac{dR(P^n_{\xi,\varkappa})}{d\lambda_1}(r) = h^n_{\varkappa,d}(r) = \frac{a_d(\varkappa)^n}{a_d(r\varkappa)} h^n_d(r),$$

where $h^n_d(r) = (d\psi^n_d/d\lambda_1)(r)$, λ_1 being Lebesgue measure in $\mathbf{R}$, and the conditional distribution of the resultant direction given the resultant length is

$$\frac{d((v./R) \mid R = r)(P^n_{\xi,\varkappa})}{d\mu_d}(w) = a_d(r\varkappa)\, e^{-r\varkappa\xi \ast w},$$

that is $v./R$ given $R = r$ follows the hyperboloid distribution (6) with direction ξ and concentration $r\varkappa$. $\square$

Proof. Omitted.

It follows from Theorem 2 that R is G- and M-sufficient for $\varkappa$ in the terminology of Barndorff-Nielsen (1978*a*).

As indicated by the distribution of R, it may be straightforward to write down the distribution of a statistics, say z, if we know the form of the lifted measure $z(\mu^n_d)$. We give a further example by stating the distribution of $T = \xi \ast v.$ and the distribution of the resultant length R given T. The statistic T will be used in the section about testing, in the situation where one wants to test a hypothesis about the direction ξ in the presence of the nuisance parameter $\varkappa$. The distributions are:

$$\frac{dT(P^n_{\xi,\varkappa})}{d\lambda_1}(t) = a_d(\varkappa)^n e^{-\varkappa t} \frac{dT(\mu^n_d)}{d\lambda_1}(t) \qquad (9\text{ a})$$

and

$$\frac{d(R \mid T = t)(P^n_{\xi,\varkappa})}{d\lambda_1}(\varkappa) = \frac{1}{r} h^n_d(r) \frac{(d-1)\pi^{(d-1)/2}}{\Gamma(d+1)/2)} \left(\left(\frac{t}{r}\right)^2 - 1\right)^{(d-3)/2} \Big/ \frac{dT(\mu^n_d)}{d\lambda_1}(t). \qquad (9\text{ b})$$

By transforming the resultant vector $v.$ to $(\cosh u, 0, \ldots, 0)'$ with a hyperbolic transformation, it is seen that $R \geqslant n$, i.e. the resultant length is greater than or equal to the number of observations. Further, since $T = \xi \ast v. = R\xi \ast (v./R)$, we find that $T \geqslant R$ (if x and y are vectors on the unit-hyperboloid then $x \ast y \geqslant 1$ with equality if and only if $x = y$, see the proof of Lemma 1).

One should note here that the concentration parameter in the conditional distribution of the resultant direction $v./R$ given the resultant length R will always be at least n times greater than the concentration parameter in the unconditional distribution of the single observation, quite contrary to the von Mises–Fisher case. This is a reflexion of the geometry of the space in which the observations lie. For the unit-hyperboloid case the observations will always in some sense cluster, so that the mean direction becomes more accurately determined.

In Theorem 2 we used $d\psi_d^n/d\lambda_1$ (see Theorem 1 for the definition of ψ_d^n). The following theorem gives the form of $d\psi_d^n/d\lambda_1$ explicitly in two and three dimensions.

Theorem 3. *For $d = 2$ we have*

$$h_2^n(r) = \frac{d\psi_2^n}{d\lambda_1}(r)$$
$$= r\pi^{n+1} Re\left\{ i^{n+1} \int_0^\infty J_0(r\tau)\,[H_0^{(1)}(\tau)]^n \tau\, d\tau \right\}$$

where J_0 is the standard Bessel function of the first kind and

$$H_0^{(1)}(\tau) = \frac{1}{\pi i}\int_{-\infty}^{\infty} e^{i\tau\cosh t}\, dt$$

is the Hankel function of the first kind.

In the case $d = 3$

$$h_3^n(r) = \frac{d\psi_3^n}{d\lambda_1}(r) = \frac{(2\pi)^{n-1}}{(n-2)!}\, r(r-n)^{n-2}, \quad r \geqslant n. \qquad \square$$

Proof. The form of h_d^n for $d = 2$ is deduced in Rukhin (1974). Let $\zeta = (1, 0, \ldots, 0)'$, $C = \zeta \ast w$ for $w \in H_d$, and

$$f_n(t) = \frac{d(\zeta \ast v.)(\mu_3^n)}{d\lambda_1}(t).$$

We now establish a relation between f_n and h_3^n. Let $s \geqslant 1$ then, from Theorem 1,

$$\mu_3^n(\zeta \ast v. \leqslant sn) = \iint_{\{\zeta\ast(v./R)=c\leqslant s,\; n\leqslant R\leqslant(sn/c)\}} d\psi_3^n\, d\mu_3$$
$$= \int_1^s \psi_3^n\left(\frac{sn}{c}\right)\frac{dC(\mu_3)}{d\lambda_1}\, d\lambda_1(c).$$

To find $dC(\mu_3)/d\lambda_1$ we use that $d\mu_3 = \sinh u\, du\, dv$, which implies

$$\mu_3(C \leqslant c) = \int_0^{\cosh^{-1}(c)}\int_0^{2\pi} \sinh u\, dv\, du = 2\pi(c-1)$$

and hence we find

$$\frac{dC(\mu_3)}{d\lambda_1}(c) = 2\pi, \quad c \geqslant 1. \tag{10}$$

Returning, we find

$$\mu_3^n(\zeta \ast v. \leqslant sn) = 2\pi\int_1^s \psi_3^n\left(\frac{sn}{c}\right) dc$$
$$= 2\pi sn \int_n^{sn} \frac{1}{x^2}\psi_3^n(x)\, dx$$

from which it follows that

$$f_n(t) = 2\pi\left(\int_n^t \frac{1}{x^2}\psi_3^n(x)\, dx + \frac{1}{t}\psi_3^n(t)\right).$$

Consequently,

$$f_n'(t) = 2\pi\frac{1}{t}h_3^n(t). \tag{11}$$

We now need to find f_n. From (10) we have $f_1(t) = 2\pi$, $t \geqslant 1$. Using $\zeta \ast v. = \sum_1^n \xi \ast v_i$ we obtain inductively

$$f_2(z) = \int_1^{z-1} f_1(z-s) f_1(s) ds = (2\pi)^2(z-2), \quad z \geqslant 2$$
$$\vdots$$
$$f_n(z) = (2\pi)^n \frac{1}{(n-1)!}(z-n)^{n-1}, \quad z \geqslant n. \tag{12}$$

Finally,

$$f_n'(z) = \frac{(2\pi)^n}{(n-2)!}(z-n)^{n-2}, \quad z \geqslant n,$$

which combined with (11) gives the desired result. $\square$

Remark. Since $T = \xi \ast v. = (A \ast \xi) \ast (A \ast v.)$ for any hyperbolic transformation A and since μ_d is invariant under hyperbolic transformations we see that $T(\mu_d^n) = (\zeta \ast v.)(\mu_d^n)$. In particular, for $d = 3$ we have

$$\frac{dT(\mu_3^n)}{d\lambda_1}(t) = f_n(t)$$

where f_n is given by (12).

Combining the results obtained in this section we find the following formulae for $d = 3$:

$$\frac{dR(P^n_{\xi,\varkappa})}{d\lambda_1}(r) = \frac{\varkappa^{n-1}}{(n-2)!}(r-n)^{n-2}e^{-\varkappa(r-n)}, \quad r \geqslant n \qquad (13)$$

i.e. $R-n \sim \Gamma(n-1), 1/\varkappa)$, and

$$\frac{dT(P^n_{\xi,\varkappa})}{d\lambda_1}(t) = \frac{\varkappa^n}{(n-1)!}(t-n)^{n-1}e^{-\varkappa(t-n)}, \quad t \geqslant n, \qquad (14)$$

i.e. $T-n \sim \Gamma(n, 1/\varkappa)$ and, finally,

$$\frac{d(R\,|\,T=t)(P^n_{\xi,\varkappa})}{d\lambda_1}(r) = (n-1)\frac{(r-n)^{n-2}}{(t-n)^{n-1}}, \quad n \leqslant r \leqslant t. \qquad (15)$$

C. *Estimation*

We shall here treat the problem of estimation within the family $\mathcal{D} = \{P_{\xi,\varkappa} \,|\, \xi \in H_d, \varkappa > 0\}$.

By direct calculation it is seen, that $\mathcal{D}$ is a regular exponential family of order d. The following results are easily proved by using the theory of regular exponential families, as described in Barndorff-Nielsen (1978*a*).

Lemma 3. *If v follows the distribution $P_{\xi,\varkappa}$ then*

$$E(v) = \frac{a_d'(\varkappa)}{a_d(\varkappa)}\xi.$$

The function $\varkappa \to a_d'(\varkappa)/a_d(\varkappa)$ is strictly decreasing and maps the open interval $(0, \infty)$ *onto* $(1, \infty)$. □

From (8) we find that

$$\frac{a_2'(\varkappa)}{a_2(\varkappa)} = \frac{-K_0'(\varkappa)}{K_0(\varkappa)} = \frac{K_1(\varkappa)}{K_0(\varkappa)} \quad \text{and} \quad \frac{a_3'(\varkappa)}{a_3(\varkappa)} = \frac{1}{\varkappa} + 1.$$

We now consider a sample $v_1, ..., v_\varkappa$ from $P_{\xi,\varkappa}$. From Lemma 3 we find that the maximum likelihood estimate of $(\xi, \varkappa)$ exists if and only if $R = R(v.) > n$ and is then given by

$$\hat{\xi} = \frac{v.}{R}, \frac{a_d'(\hat{\varkappa})}{a_d(\hat{\varkappa})} = \frac{R}{n}.$$

In the same way as it was seen that $R \geqslant n$, it may be seen that equality $R = n$ holds if and only if the observations are identical, $v_1 = ... = v_n$. Since the event $\{v_1 = ... = v_n\}$ has probability zero, we have that the maximum likelihood estimate of $(\xi, \varkappa)$ exists with probability one provided $n > 1$.

As noted afer Theorem 2, R is G- and M-sufficient for $\varkappa$, so there may be strong reasons for estimating $\varkappa$ in the marginal distribution of R. The marginal estimate $\tilde{\varkappa}$ is the unique solution of

$$\frac{a_d'(\varkappa)}{a_d(\varkappa)} = \frac{R}{n}\frac{a_d'(R\varkappa)}{a_d(R\varkappa)}$$

and $\tilde{\varkappa} < \hat{\varkappa}$. The uniqueness may be proved as in the von Mises–Fisher case (Schou, 1978) using Theorem A.1 in Jørgensen (1980). The relevance of estimating in the marginal distribution of R can be illustrated by the work of Schou (1978) for the von Mises–Fisher case.

In three dimensions we find the following explicit formulae

$$\hat{\varkappa} = \frac{1}{R/n - 1} \quad \text{and} \quad \tilde{\varkappa} = \frac{1 - 1/n}{R/n - 1}.$$

Since R is invariant under hyperbolic transformations, it is clear that $\hat{\varkappa}$ is ancillary for fixed $\varkappa$, i.e. the distribution of $\hat{\varkappa}$ under $P_{\xi,\varkappa}$ does not depend on the parameter ξ. This has been used in three dimensions for an example of conditionality resolution by Barndorff-Nielsen (1980).

D. *Testing*

The test theory for the distribution $P_{\xi,\varkappa}$ has many points of resemblance with the theory for the von Mises–Fisher distribution. We shall therefore limit ourselves to mentioning a few results.

We first discuss the single sample. Since R is G- and M-sufficient for $\varkappa$ a test for $\varkappa \leqslant \varkappa_0$ may be performed in the marginal distribution of R, small values of R being critical. Next, let us consider a test for $\xi = \xi_0$, supposing $\varkappa > 0$ unknown. From the joint distribution of $v_1, ..., v_n$ and the distribution of T, given in (9a), it is seen that under the hypothesis $\xi = \xi_0$ the statistic $T = \xi_0 \ast v.$ is sufficient and complete. Accordingly, if we want to perform a similar test this must be done in the conditional distribution of $v.$ given $T = t$. This conditional distribution is easily seen to depend on $v.$ through (R, t) only so that the test may be performed in the conditional distribution of R given $T = t$, see (9b), with critical values of R close to n $(n \leqslant R \leqslant T)$. In the von Mises–Fisher case this test has a serious drawback, since it may accept observations that have a resultant direction opposite to ξ_0. In the hyperbolic case this problem is not present; once again, this is related to the geometry of the spaces involved. The point may be illustrated in two dimensions as shown in Fig. 1.

From (15) it is seen that the level of significance in three dimensions is

$$\left(\frac{r-n}{t-n}\right)^{n-1}. \qquad (16)$$

As a curious fact, one obtains the same result by applying Fisher's fiducial argument. The fiducial argument for the von Mises–Fisher distribution in three dimensions (Fisher, 1953) may be carried over

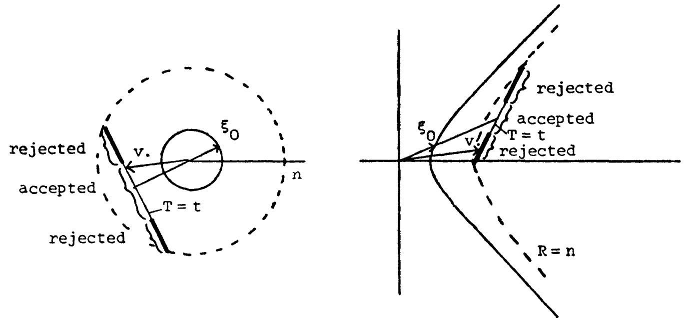

Fig. 1. Critical regions for the hypothesis $\xi=\xi_0$. (*a*) von Mises model; (*b*) hyperbola model.

to the distribution $P_{\xi,\varkappa}$ without change, in fact the calculations are easier for the distribution $P_{\xi,\varkappa}$. The argument is as follows. The test for $\xi=\xi_0$ should be performed in the conditional distribution of C given $R=r$, where

$$C=\frac{T}{R}=\xi_0 * \frac{v.}{R},$$

since, for fixed $\varkappa$, R is ancillary and the conditional distribution of $v.$ given R is, under the hypothesis $\xi=\xi_0$, a function of C alone. From (13), (14) and (15) it is seen that

$$\frac{d(C\,|\,R=r)\,(P^n_{\xi_0,\varkappa})}{d\lambda_1}(c)=r\varkappa\, e^{-\varkappa r(c-1)}. \tag{17}$$

Now this depends on the unknown $\varkappa$. To eliminate $\varkappa$ a fiducial density is derived from the marginal distribution of R (recall that R is G- and M-sufficient for $\varkappa$ in the family $\mathcal{D}$); from (13) the density is found to be

$$\frac{(r-n)^{n-1}}{(n-2)!}\varkappa^{n-2}e^{-(r-n)\varkappa}.$$

Finally, integrating out $\varkappa$ in (17), we find the following density for C given $R=r$

$$(n-1)\,r\frac{(r-n)^{n-1}}{(rc-n)^n}.$$

Since large values of C are critical for $\xi=\xi_0$ we find the level of significance to be (16).

To make the discussion complete it should be noticed that the inference for ξ based on the fiducial distribution for c is the same as the inference based on the pivotal quantity $(T-n)/(R-n)$ (see (13) and (14)), just as the inference for the mean of a normal distribution.

Also in analogy to the von Mises–Fisher case, approximate tests may be constructed on the basis of the asymptotic normality of the distribution (6) as $\varkappa\to\infty$, discussed in section 3.A. We consider again the hypothesis $\xi=\xi_0$. Then, under the hypothesis, the variate

$$\frac{(T-R)}{(R-n)}(n-1)$$

tends to a F-distribution with $d-1$ and $(n-1)(d-1)$ degrees of freedom as $\varkappa\to\infty$. The statistic $((T-R)/(R-n))(n-1)$ may therefore be used for an approximate test, large values being critical (for $d=3$ the test is, in fact, exact).

Next, suppose several samples are available. Let the number of samples be q, i.e. $v_{ij}\sim P_{\xi_i,\varkappa_i}$, $i=1,\dots,q$, $j=1,\dots,n_i$. Set $R_i=R(v_{i.})$, $n=\sum_1^q n_i$, and $R=R(v..)$. First we consider an asymptotic test for the hypothesis of a common precision, $\varkappa_1=\dots=\varkappa_q$. In analogy to the von Mises–Fisher case, we find that the variate $2\varkappa_i(R_i-n_i)$ tends to a chi-squared distribution with $(n_i-1)(d-1)$ degrees of freedom as $\varkappa\to\infty$, which may be used to construct a test. Specifically, a test of identical scale parameter for gamma distributed variables may be applied (again, for $d=3$ the test is exact, cf. (13))

Supposing now that $\varkappa_1=\dots=\varkappa_q$ we consider the hypothesis of a common mean direction $\xi_1=\dots=\xi_q$. As in the von Mises–Fisher case a test may be constructed from the distribution of $(R_1,\dots,R_q)$ given the total resultant length R. This distribution is independent of the parameters involved, as may be learned from the last part of Theorem 1. In three dimensions the conditional distribution of $(R_1,\dots,R_q)$ given R has density

$$\frac{(n-2)!\,(R-\sum_1^q R_i)^{q-2}}{(q-2)!\,(R-n)^{n-2}}\prod_1^q\frac{(R_i-n_i)^{n_i-2}}{(n_i-2)!},$$

$$R_i\geq n_i,\ \sum_1^q R_i\leq R. \tag{18}$$

This may be shown as follows. Notice first that for $w_i\in H_3$, $\beta_i\geq 1$ we have

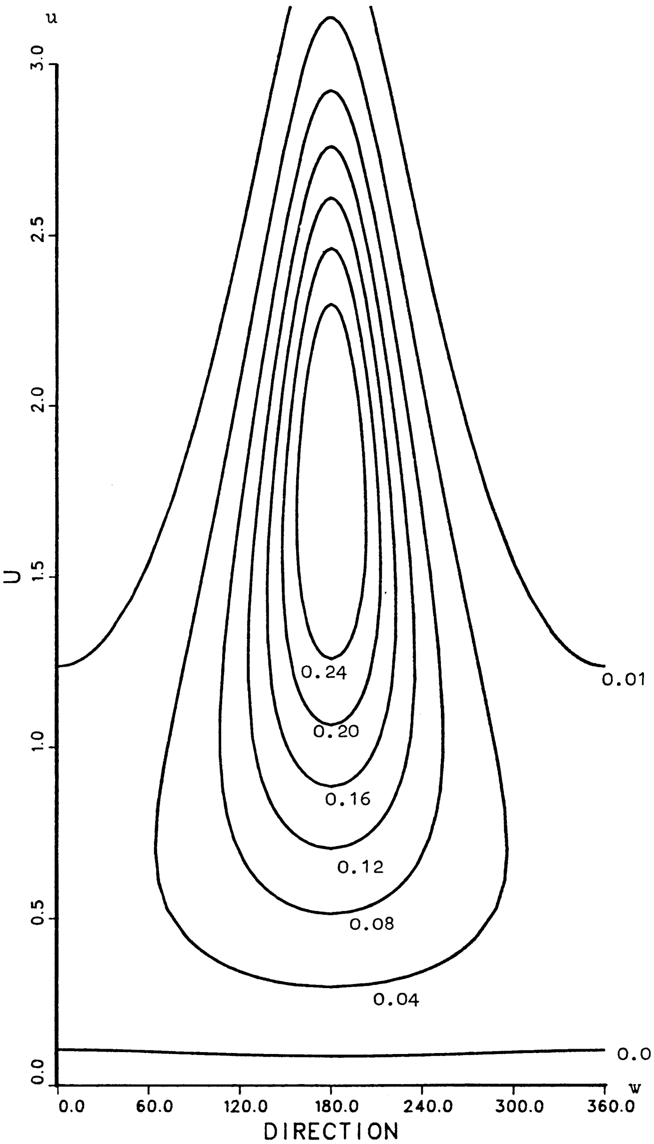

Fig. 2. Contour plots for the hyperboloid distribution with parameters $\varkappa=1$, $\sinh\chi=1$, $\theta=180°$.

$$\left(\frac{\sum\beta_i w_i}{R(\sum\beta_i w_i)}, R(\sum\beta_i w_i)\right)(\mu_3^q)=\mu_3\times\phi_q,$$

for some measure ϕ_q. Using the line of argument of Theorem 3 we find that

$$\frac{d\phi_q}{d\lambda_1}(t)=\frac{(2\pi)^{q-1}}{\beta_1\cdots\beta_q(q-2)!}t(t-\sum\beta_i)^{q-2},\ t\geqslant\sum\beta_i \tag{19}$$

Now, the rest of the proof is identical to the proof for the Fisher distribution, which may be seen in Mardia (1972), section 8.6. The simplicity of (18) and (19) should be compared to the more complicated formulae (8.6.39) and (8.6.17) of Mardia (1972). Finally an asymptotic test of the hypothesis $\xi_1=\ldots=\xi_q$ may be based on the variate

$$\frac{(R-\sum_i^q R_i)/(q-1)}{(\sum_i^q R_i-n)/(n-q)}$$

which tends to a F-distribution with $(d-1)(q-1)$ and $(d-1)(n-q)$ degrees of freedom as $\varkappa\to\infty$, and large value are critical for the hypothesis.

4. A data example

In this section we present some data for which the hyperboloid distribution provides a suitable descrip-

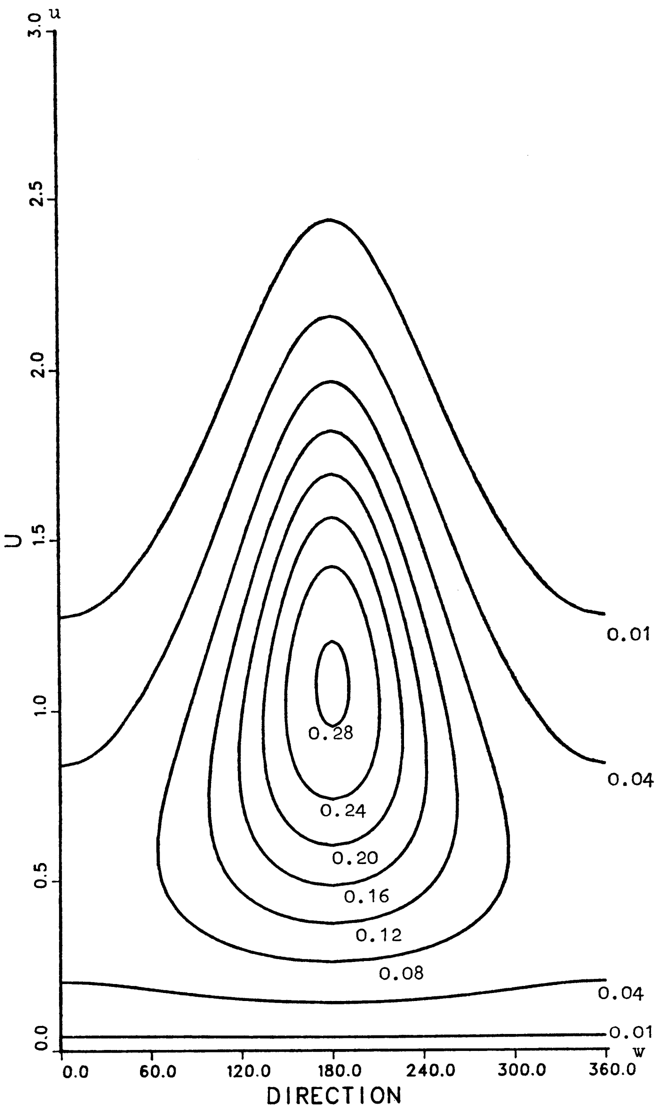

Fig. 3. Contour plots for the hyperbolorid distribution with parameters $\varkappa=2$, $\sinh\chi=0.5$, $\theta=180°$.

tion. Recall that in three dimension the hyperboloid distribution has the form

$$dP_{\chi,\theta,\varkappa}(u,w)=\frac{\varkappa e^{\varkappa}}{2\pi}\sinh u$$

$$\times\exp(-\varkappa(\cosh\chi\cosh u-\sinh\chi\sinh u\cos(w-\theta)))$$

$$\times\,dw\,du\quad u\geqslant 0,\ w\in[0,2\pi). \tag{20}$$

To get a feeling for this distribution some contour-plots have been drawn, see Figs. 2 and 3 where the numbers on the contour lines are the values of the density (20).

The data we shall consider consist of simultaneous records of wind speed and wind direction collected over the period 1958–1967 at Risø in Denmark. For the present study we have chosen to discuss the measurements of the direction and speed of the wind taken at noon on certain days in the months of November and December from the ten year period mentioned. Data sets 1 and 2 are from dates 3, 7, 11, 15, 19, 23, 27 and 30 and were recorded, respectively, at 7 m and 56 m above the ground level, while set no. 3 was taken on the dates 1, 5, 9, 13, 17, 21, 25, 29 at 7 m above ground level; see Fig. 4. The unit of the speed measurements is m/s and directions are measured in degrees 0–360°.

As mentioned in section 3.A the conditional distribution of w given u is von Mises with mean direction θ and concentration $\varkappa\sinh\chi\sinh u$.

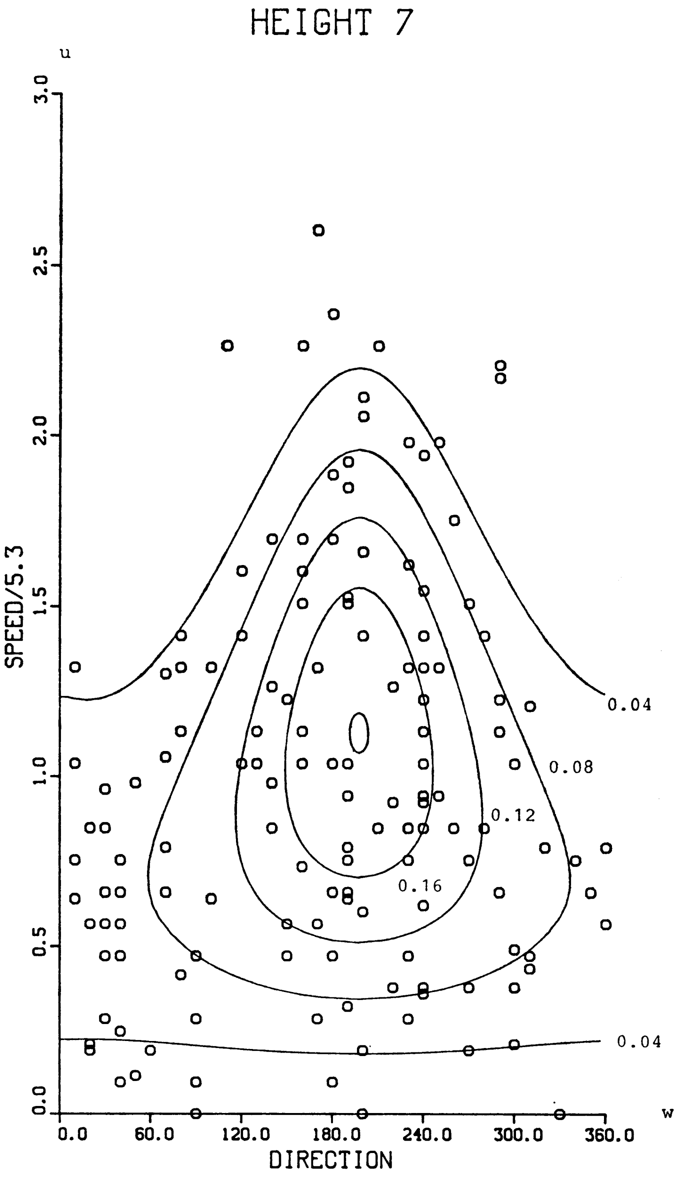

Fig. 4. **Wind data, set no. 1. Observed points and contour plots for the fitted hyperboloid distribution.**

Accordingly, the data were divided with respect to speed into a number of groups, the ith group being $2(i-1) \leqslant \text{speed} < 2i$, and a von Mises distribution was fitted within each group. The estimated mean direction and concentration for data-set 1 are shown on Fig. 5. The numbers on the directions are the group numbers, and the vertical lines in the coordinate system give the value of the concentration; the curve in this drawings will be explained later.

In order to be able to compare the different data sets the speed was divided by a mean speed for the period considered and at the given height (these numbers were obtained from Petersen (1975) and found to be 5.3 m/s and 7.2 m/s for the heights 7 m and 56 m, respectively). In this way the transformed observations do not depend on the particular scale chosen for the measurement of the speed. This could of course also be effectuated by introducing a scale parameter σ for u in the model; for this extended model the partially maximized likelihood function for σ, i.e. the likelihood function maximised for fixed σ, is very flat around the maximum and, in fact, the above mentioned procedure gives a value of the likelihood function which is very close to the maximum obtained by introducing a scale parameter.

Table 1. *The maximum likelihood estimates of the parameters in the hyperboloid distribution*

Data set no.	$\hat{\varkappa}$	$\sinh \hat{\chi}$	$\hat{\theta}$
1	1.4417	0.3584	197.5°
2	1.3870	0.2889	195.6°
3	1.2978	0.3430	190.3°

Next, the model was fitted to the transformed data by finding the maximum likelihood estimates of the parameters.

Three ways have been used to decide whether the fit is acceptable. First the contour-plots of the fitted distribution have been compared with the data, for dataset 1 see Fig. 4; the numbers on the contour lines are the values of the density (20). Secondly, the curve for the concentration in the conditional distribution of the direction given the speed, i.e. $\varkappa \sinh \chi \sinh u$ as a function of u, has been compared with the estimated values within each group; in Fig. 5 the curve drawn is $\hat{\varkappa} \sinh \hat{\chi} \sinh u$ where $\hat{\varkappa}$ and $\hat{\chi}$ are the maximum likelihood estimates. And finally a drawing of a histogram for the marginal distribution of the speed together with the marginal density of the fitted distribution have been made, see Fig. 6.

To illustrate the previous theory we now perform some tests.

It is obvious from the look of the plots in Fig. 4, that the directions and the speed are not independent. According to this the hypothesis $\chi = 0$ (equivalent to $\xi = (1, 0, 0)$ in the previous notation), which means that the direction and speed are independent with the direction being uniformly distributed, should be rejected. So, not very surprisingly, using the test described in section 3.D gives a level of significance less than 10^{-6} for all three data sets.

Next, since data sets no. 1 and 3 were taken in the same period and the same height but on different

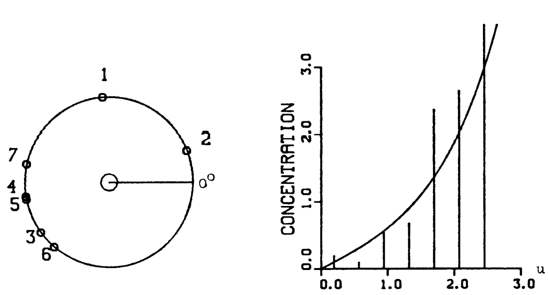

Fig. 5. Wind data, set no. 1. Estimated mean direction and concentration within each group. The curve is $\hat{\varkappa} \sinh \hat{\chi} \sinh u$.

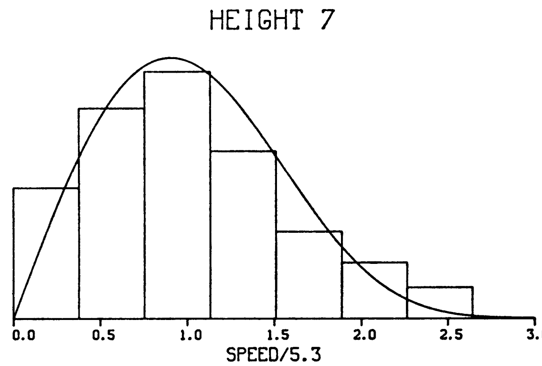

Fig. 6. Data set no. 1. Marginal distribution for u. Full line: fitted hyperboloid distribution.

days they should be very similar. Using $R_i - n_i \sim \Gamma(n_i - 1, 1/\varkappa_i)$ (see (13)) one may first test $\varkappa_1 = \varkappa_2$, which gives a level of significance equal to 0.83. Under the hypothesis $\varkappa_1 = \varkappa_2$ a test of $\xi_1 = \xi_2$ may be performed in the conditional distribution of (R_1, R_2) given R, the resultant length of the total sample. For instance, one may use $R_1 + R_2$ given R with small values as critical and the level of significance is then found from (18) to be

$$\left(\frac{R_1 + R_2 - n}{R - n}\right)^{n-2},$$

which in our case gives 0.85.

Finally, we compare data sets 2 and 3. They consist of data from the same period but from different days and different heights. Moreover, it should be kept in mind that we have rescaled the speed measurements to obtain mean speed 1 for each data set. Following the same procedure as above, the test of $\varkappa_1 = \varkappa_2$ gives a level of significance equal to 0.33, and the test of $\xi_1 = \xi_2$ under $\varkappa_1 = \varkappa_2$ has level of significance equal to 0.75.

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